

B.Sc. BOTANY — YEAR I

MAJOR COURSE III

# **APPLIED BOTANY**

UNIT V — Plant Tissue Culture, Recombinant DNA and Bioinformatics

## **Detailed Teacher-Style Notes**

Definitions • Concepts • Tables • Flow diagrams • Examples • Applications • Exam points

Prepared in the same detailed format as the Major Course II notes.

# UNIT V — BIOTECHNOLOGY AND BIOINFORMATICS

Syllabus topics covered: plant tissue culture—definition, types and importance; recombinant technique—introduction, tools, technique and importance; bioinformatics—definition, concept, tools; introduction to BLAST and FASTA; importance of bioinformatics; Indian scientists' contribution in plant tissue culture; applications of BLAST and FASTA.

## 5.1 Plant tissue culture

**Definition — Plant tissue culture:** The aseptic in-vitro culture of plant cells, tissues or organs on a defined nutrient medium under controlled environmental conditions.

Scientific basis: plant cells can show **totipotency**, meaning that under appropriate conditions some living plant cells can regenerate a complete plant. Actual regeneration depends on genotype, explant, medium and culture conditions.

Standard plant tissue-culture workflow



### Types / major culture forms

| Type                      | Core idea                            | Use                                             |
|---------------------------|--------------------------------------|-------------------------------------------------|
| Callus culture            | Unorganized cell mass grown in vitro | Regeneration and experimental studies           |
| Organ culture             | Culture of organs or organ parts     | Developmental and propagation studies           |
| Meristem culture          | Culture of shoot-tip/meristem tissue | Rapid multiplication and clean stock programmes |
| Embryo culture            | Culture of isolated embryo           | Rescue and developmental studies                |
| Cell suspension culture   | Cells grown in liquid medium         | Mass culture and metabolite studies             |
| Anther/microspore culture | Culture of male reproductive tissues | Haploid and doubled-haploid work                |

### Importance

- Rapid clonal multiplication of selected plants.
- Production of uniform planting material under controlled conditions.
- Conservation and propagation of valuable germplasm.
- Support for breeding, transformation and molecular studies.
- Production of specialized metabolites in selected culture systems.

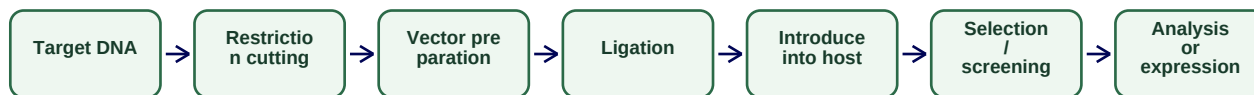
## 5.2 Recombinant DNA technique

**Definition — Recombinant DNA technology:** A set of molecular methods used to construct and work with DNA molecules formed by joining genetic material from different sources or by designed manipulation of DNA sequences.

### Core tools

| Tool                                 | Role                                                          |
|--------------------------------------|---------------------------------------------------------------|
| Restriction endonucleases            | Cut DNA at specific recognition sequences                     |
| DNA ligase                           | Joins compatible DNA fragments                                |
| Vectors                              | Carry inserted DNA into a host system                         |
| Host cells                           | Replicate or express introduced genetic material              |
| Polymerases and related enzymes      | Support DNA synthesis and amplification in relevant workflows |
| Selectable markers / screening tools | Help identify cells carrying the desired construct            |

### Simplified recombinant-DNA workflow



### Importance

- Gene-function studies.
- Production of recombinant biomolecules.
- Plant genetic improvement research.
- Molecular diagnostics and biotechnology.
- Foundation for many modern genetic engineering applications.

## 5.3 Bioinformatics

**Definition — Bioinformatics:** An interdisciplinary field that uses computational methods, databases and algorithms to store, retrieve, analyze and interpret biological information such as DNA, RNA and protein sequences.

### Bioinformatics pipeline



### Common tool categories

| Category                  | Example function                                        |
|---------------------------|---------------------------------------------------------|
| Sequence databases        | Store and retrieve nucleotide/protein sequences         |
| Sequence similarity tools | Compare a query sequence with database sequences        |
| Alignment tools           | Identify similarities and differences between sequences |
| Genome browsers           | Explore genomic regions and annotations                 |
| Phylogenetic tools        | Infer relationships from sequence data                  |
| Visualization tools       | Display sequences, structures or relationships          |

## 5.4 BLAST and FASTA

**Definition — BLAST:** Basic Local Alignment Search Tool; a widely used family of programs for finding local sequence similarity between a query sequence and sequences in a database.

**Definition — FASTA:** A sequence-comparison method and, in common usage, a simple text-based sequence format in which a header line begins with the character > followed by sequence data.

| Feature       | BLAST                                      | FASTA / FASTA-style search concept                        |
|---------------|--------------------------------------------|-----------------------------------------------------------|
| Main purpose  | Rapid local similarity search              | Sequence representation and similarity-search framework   |
| Typical input | Nucleotide or protein query                | Nucleotide or protein sequence                            |
| Main output   | Database matches with alignment statistics | Similarity matches/alignments depending on implementation |
| Use           | Identify homologous or similar sequences   | Store/compare sequence information                        |

### Applications of BLAST and FASTA

- Identify similar sequences and possible homologs.
- Check whether a sequence resembles known genes or proteins.
- Support annotation of newly obtained sequences.
- Compare plant, microbial or other biological sequences.
- Generate hypotheses for further experimental validation.

## 5.5 Indian scientists' contribution in plant tissue culture

The syllabus explicitly asks for this activity. For a final submitted answer, use contributions supported by your prescribed textbook/teacher or a verified institutional source. The syllabus page itself does not name specific scientists, so unsupported names or achievements should not be invented in the notes.

**Exam focus:** Unit V high-yield: definition and workflow of plant tissue culture; totipotency; culture types; recombinant-DNA tools and steps; bioinformatics definition; BLAST vs FASTA comparison and applications.